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Properties of Materials
Lecture 4

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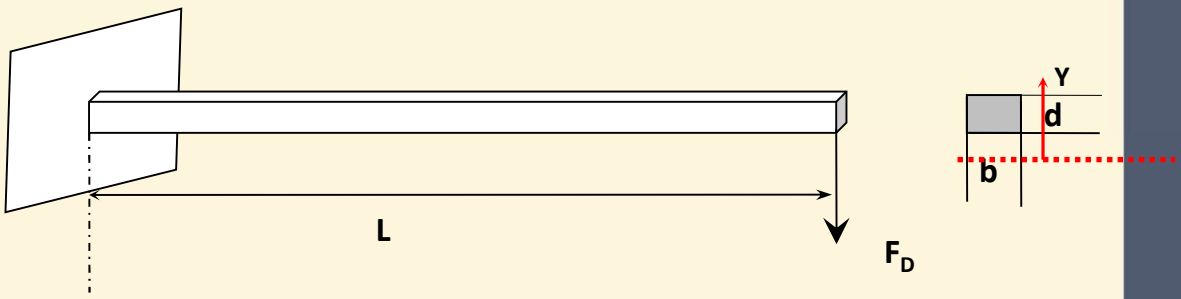
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Lecture 2: Concepts and Terminology (1), Properties of Engineering Materials (2), Mechanical Properties (1), Elastic Properties, Elastic Properties: Covalent and Metallic Bonds - Ionic Model - The Origin of Modulus, Plastic Properties, Fracture / Fatigue / Creep, Thermal Properties (1), Density and Corrosion, Corrosion: Anodic and Cathodic Reactions, Ellingham Diagram
Lecture 3: Structure-Property-Processing Relationship, Structure Sensitive Properties, Structure Insensitive Properties, The Three-Part Relationship, Materials Property Web Sites, Materials Selection, Material Property - Charts: Modulus - Density, Mechanical Properties (2), Thermal Properties (2), A TOOL for Materials Selection, Structure Sensitivity, Structure Sensitivity in Metals, Controlling Structure, Case Study
Lecture 4: Availability of Materials, Case Study, Exam Question 2013, Example Questions, The World of Materials (2), Materials Selection in Design (2), Properties of Engineering Materials (3), Three-Way Relationship (2), Learning Objectives, Why Do We Need a Selection Methodology, Mechanical Properties (3), Thermal Properties (3), Trade-Offs Must Be Made, Materials Selection in Design (3), Materials Selection Process (1)

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Availability of materials

Case study



For a given material the material index for strong light beam :

$$\frac{\sqrt{\sigma_f}}{\rho} = C$$

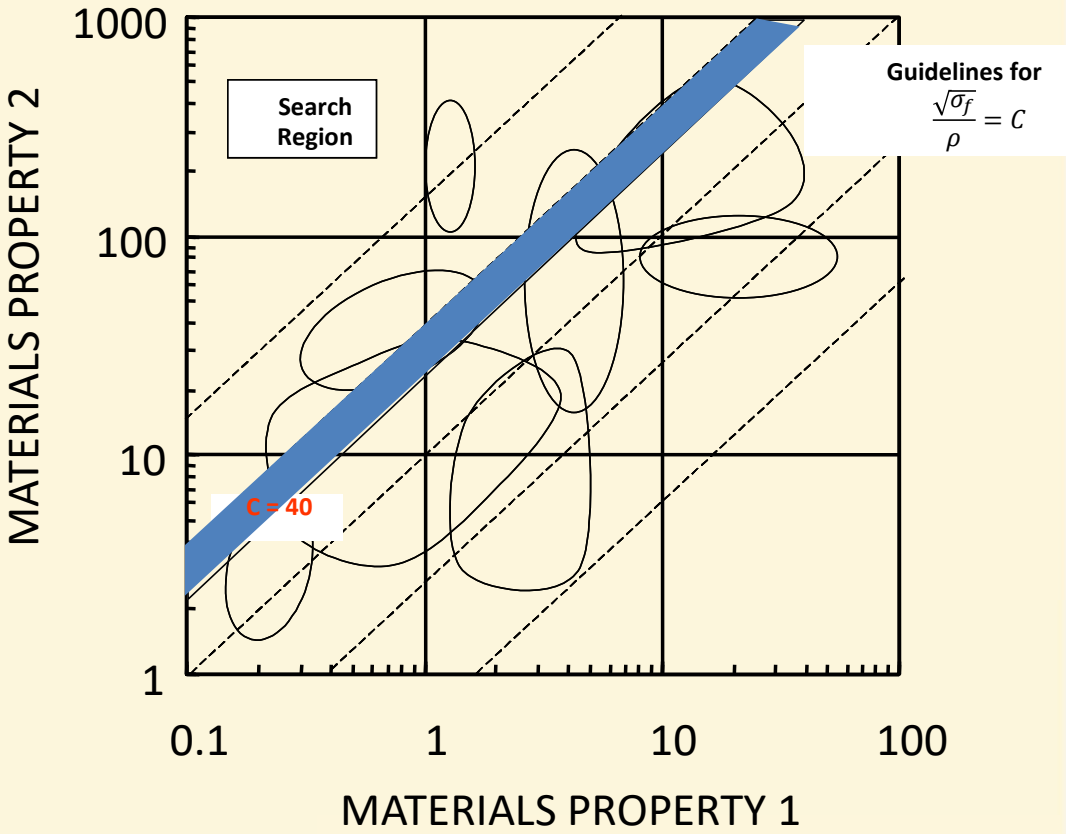
Taking logs gives the expression:

$$\frac{1}{2}\log_{10}\sigma_f - \log_{10}\rho = \log_{10} C$$

Which is of the form

$$y = mx + c$$

So we can now plot $\log_{10}\sigma_f$ versus $\log_{10}\rho$ and plot guidelines with gradient m = 2



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Case study

$$\frac{\sqrt{\sigma_f}}{\rho} = C \qquad \frac{1}{2}\log_{10}\sigma_f - \log_{10}\rho = \log_{10} C$$

Low alloy steels

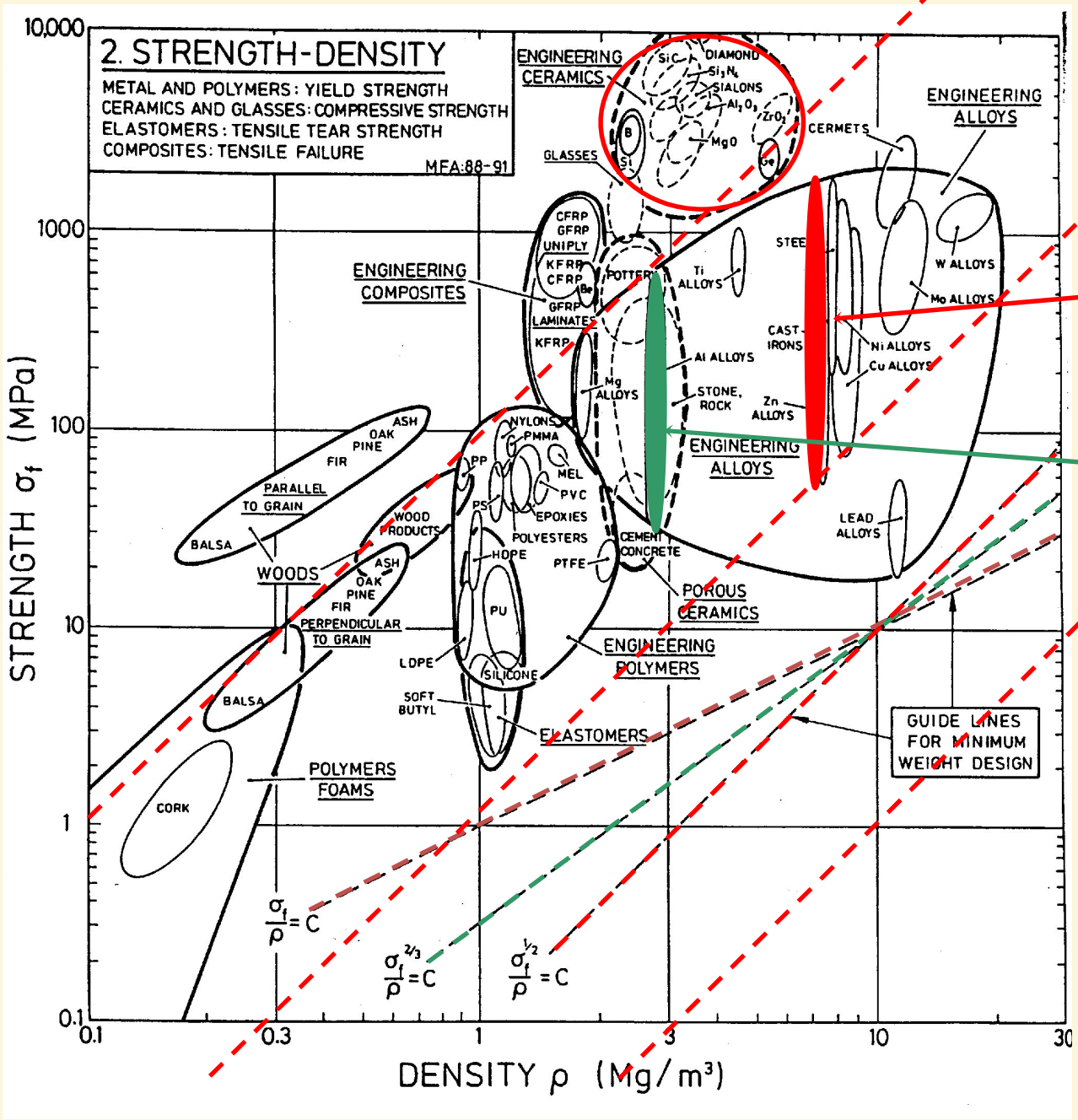
ρ: 7.8-7.85 Mgm^{-3}

σ: 50-1980 MPa

Al alloys

ρ: 2.52-2.54Mgm^{-3}

σ: 40-510 MPa



C=10

C=1

low alloy steels

Al alloys

C=0.1

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Exam question 2013

At the Battle of Bosworth in 1485, the best armour was almost impervious to arrows. Arrow shafts are designed to be light and stiff so they can be fired over a long distance. Historically, the shaft of an arrow was made from wood, often poplar but many other woods were used. You are required to compare and contrast modern materials to form the shaft of an arrow. The arrow shaft is a rod of length L which is fixed for any given archer. The mass of the arrow is critical since a lighter arrow can be fired further, however, at the release of the arrow and the impact in its target, a high force is experienced and buckling must be resisted if the arrow is to find its target with accuracy and inflict damage. The elastic buckling load of a column of length L and radius r is

$$F_{crit} = \frac{\pi^2 EI}{L^2}$$

where E is the modulus and I is the second moment of the area of the column given by

$$I = \frac{\pi r^4}{4}$$

Derive a suitable merit index which would allow you to minimise the weight of the arrow subject to the arrow shaft not buckling.

It is also desirable that the shaft should be as narrow as possible, as thicker arrows will exhibit greater air resistance. Derive a second merit index where the arrow should be as thin as possible without buckling.

Exam question 2013: Answer

The requirement is to minimise the mass

Mass of a shaft is given by;

$$m = \pi r^2 L \rho \qquad (1)$$

Elastic buckling load of a column of length L and radius r is;

$$F_{crit} = \frac{\pi^2 EI}{L^2} \qquad (2)$$

Where I is the second moment of area;

$$I = \frac{\pi r^4}{4} \qquad (3)$$

The load must not exceed F_{crit}. Insert (3) into (2)

$$F \leq \frac{\pi^3 E r^4}{4L^2} \qquad (4)$$

Exam question 2013: Answer

If we rearrange equation (1) and square it we get an expression…..

$$r^4 = \frac{m^2}{\pi^2 L^2 \rho^2} \qquad (5)$$

Insert (5) into (4)

$$F \leq \frac{\pi^3 E m^2}{4L^4 \pi^2 \rho^2}$$

Rearranging for m, which we wish to minimise. If the arrow is not going to buckle then,

$$m \geq \left(\frac{4F}{\pi}\right)^{\frac{1}{2}} L^2 \left[\frac{\rho}{E^{\frac{1}{2}}}\right]$$

Design

Materials properties

Exam question 2013: Answer

All materials properties in square brackets. So if we want to minimise m then we need to maximise;

$$M_1 = \frac{E^{\frac{1}{2}}}{\rho}$$

To minimise the cross sectional area of the shaft we simply need to rearrange

$$F \leq \frac{\pi^3 E r^4}{4L^2}$$

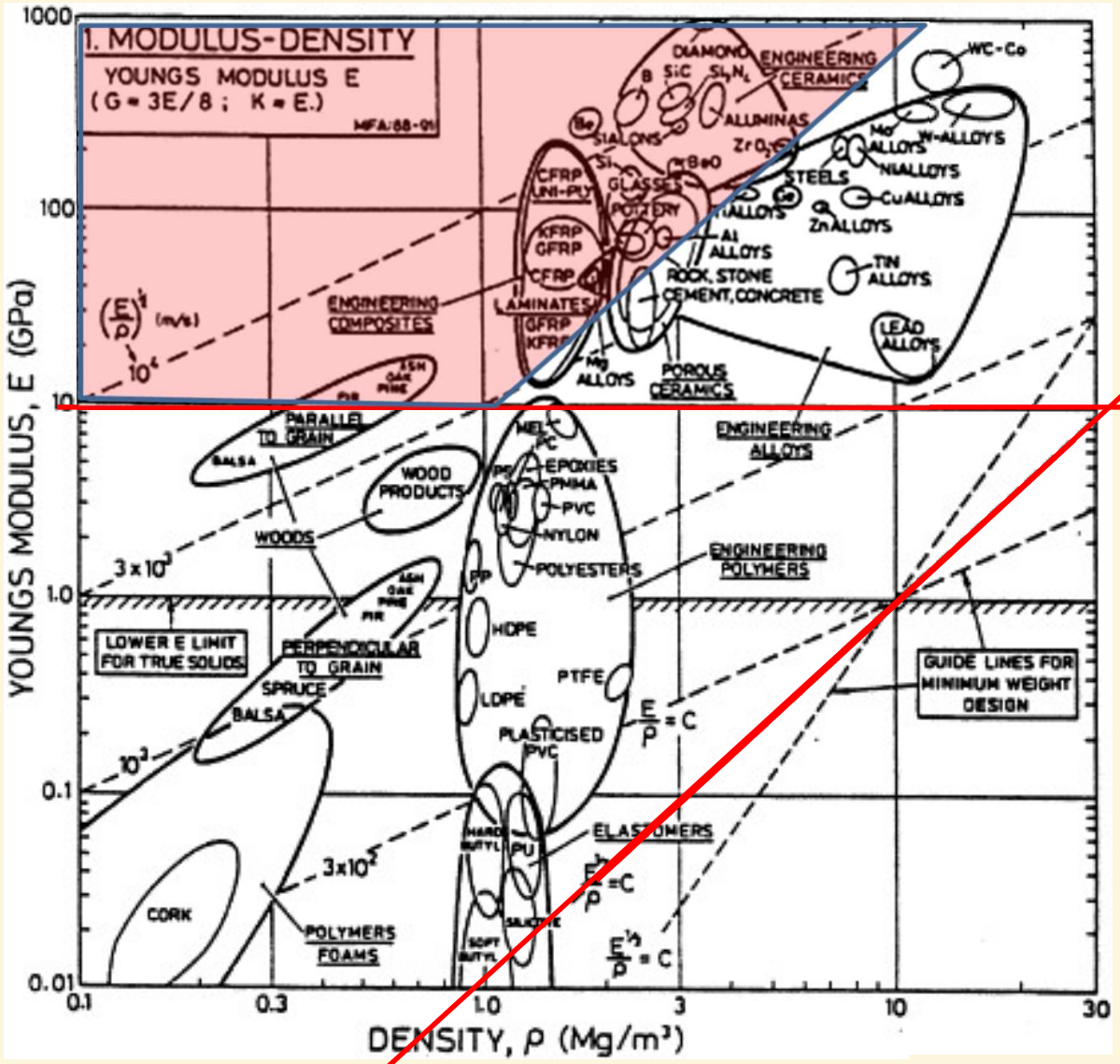
$$r \geq \left(\frac{4F}{\pi^3}\right)^{\frac{1}{4}} (L)^{\frac{1}{2}} \left[\frac{1}{E}\right]^{\frac{1}{4}}$$

Again, materials properties are in square brackets. To minimise r we should maximise;

$$M_2 = E$$

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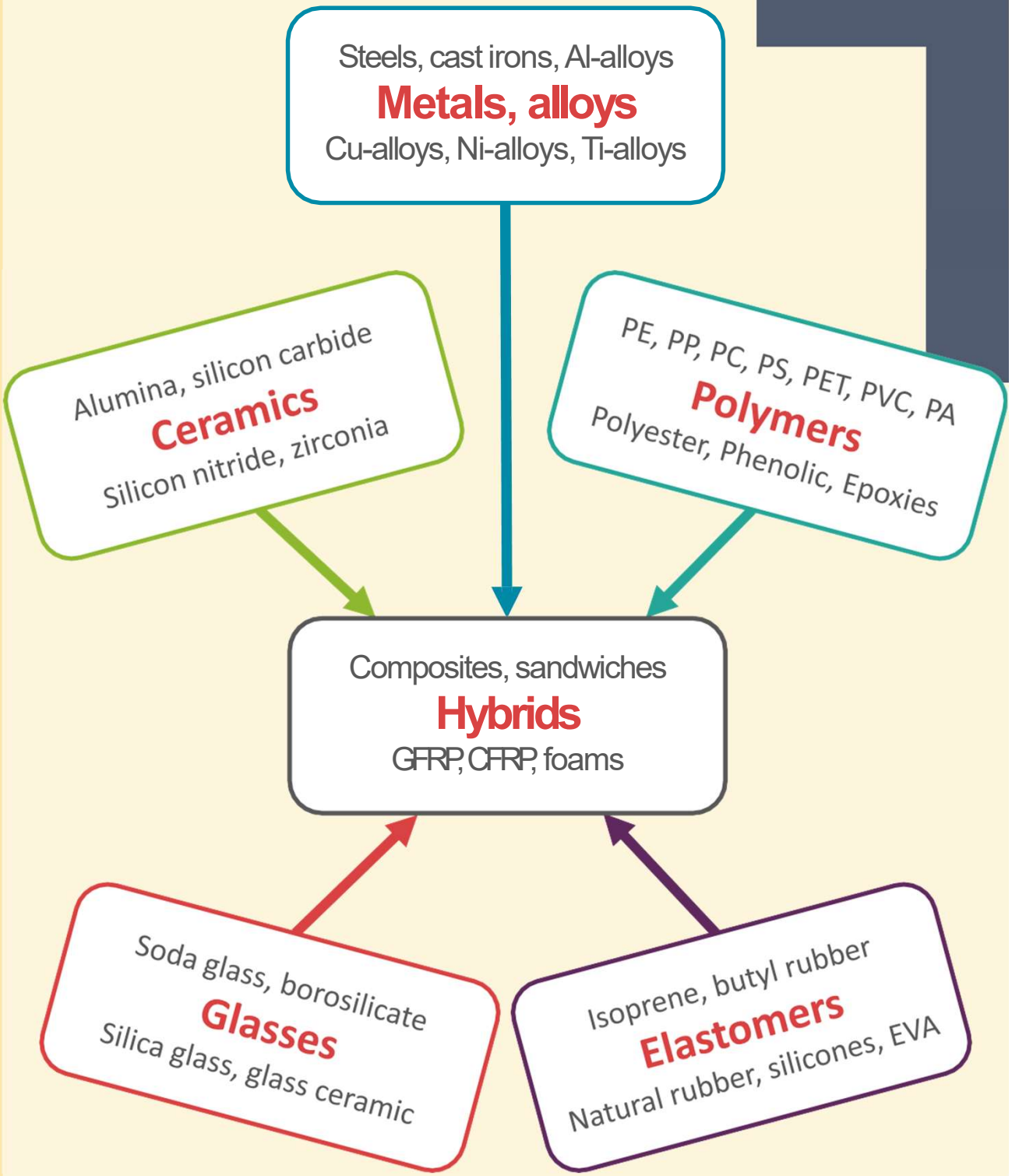
Exam question 2013: Answer



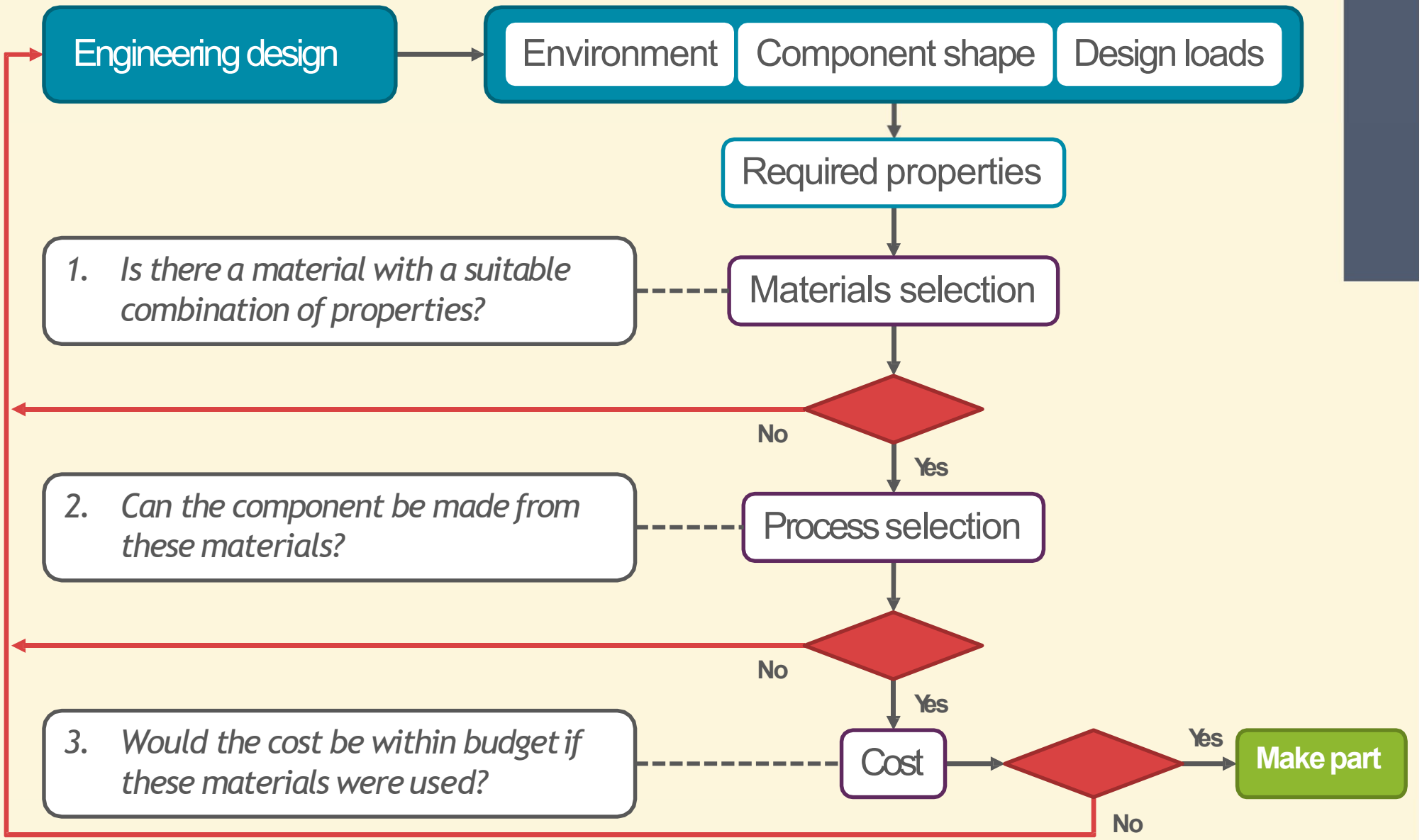
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The World of Materials

- >150,000 different materials available
- 5 families of materials
- Hybrids combine two or more of the others

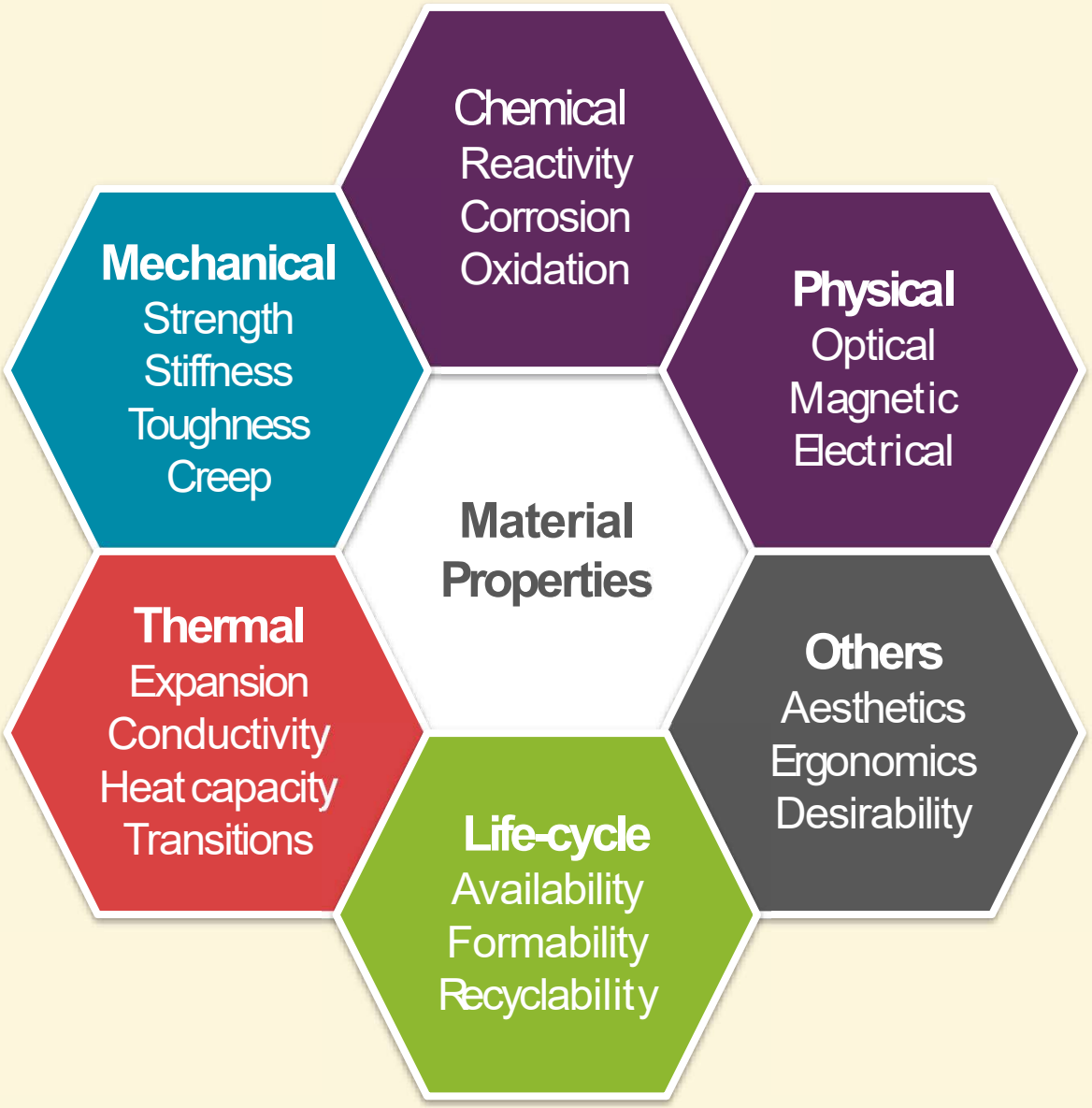


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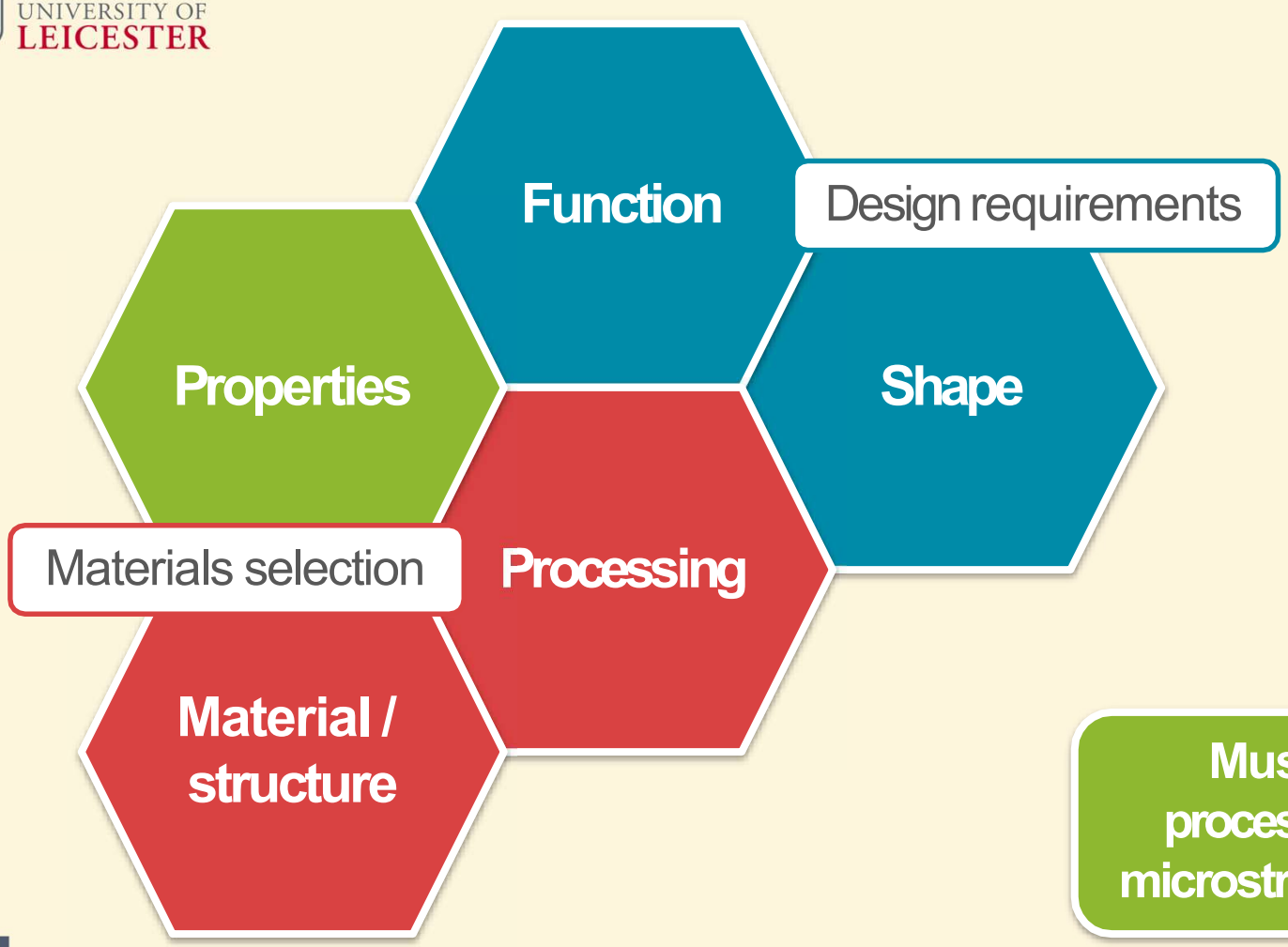
Materials selection in design

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Properties of engineering materials

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Must understand how processing history controls microstructures and properties

Three-way relationship

- Processing: shaping and tailoring material properties
- Structure: controlled by composition and processing
- Properties: controlled by microstructure

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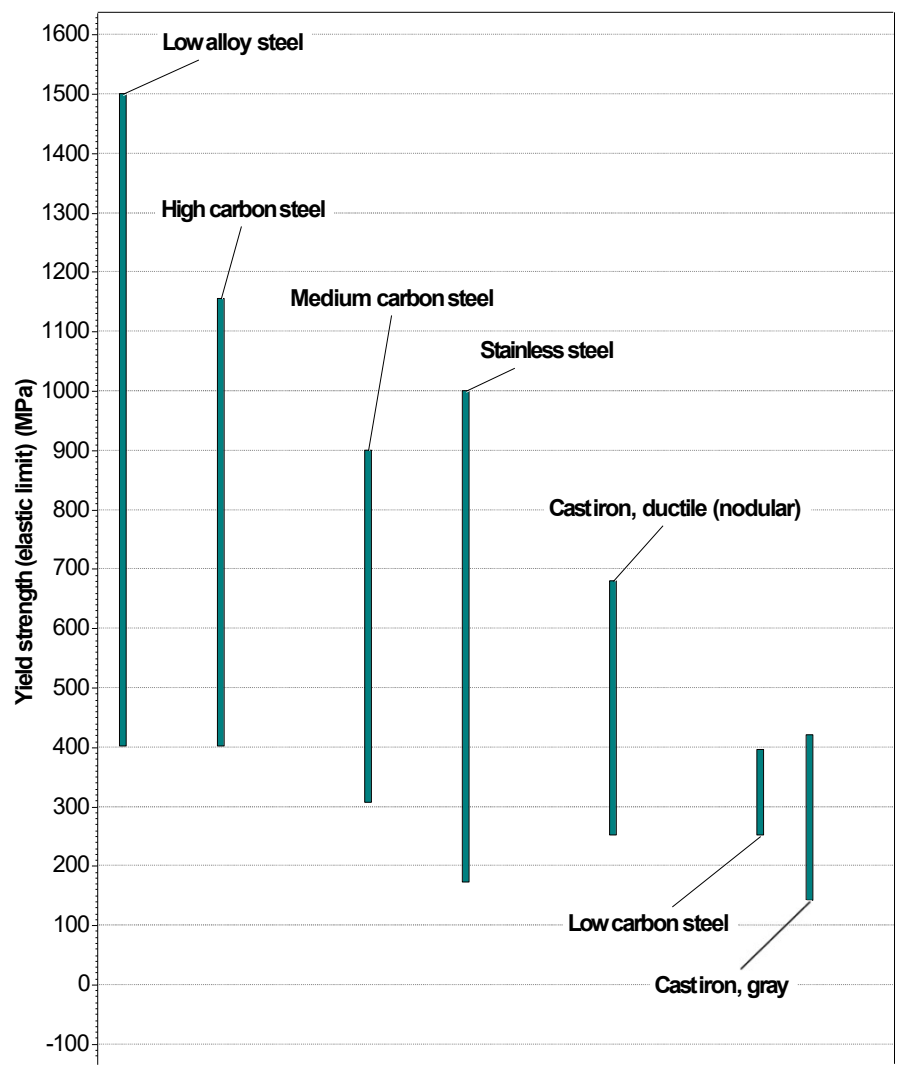
Learning objectives

- To develop a methodology for selecting materials in Engineering Design:
  - **Translating** design requirements
  - **Screening** materials
  - **Ranking** materials based on derived **performance indices**
  - **Refining** our selection based on **supporting information**

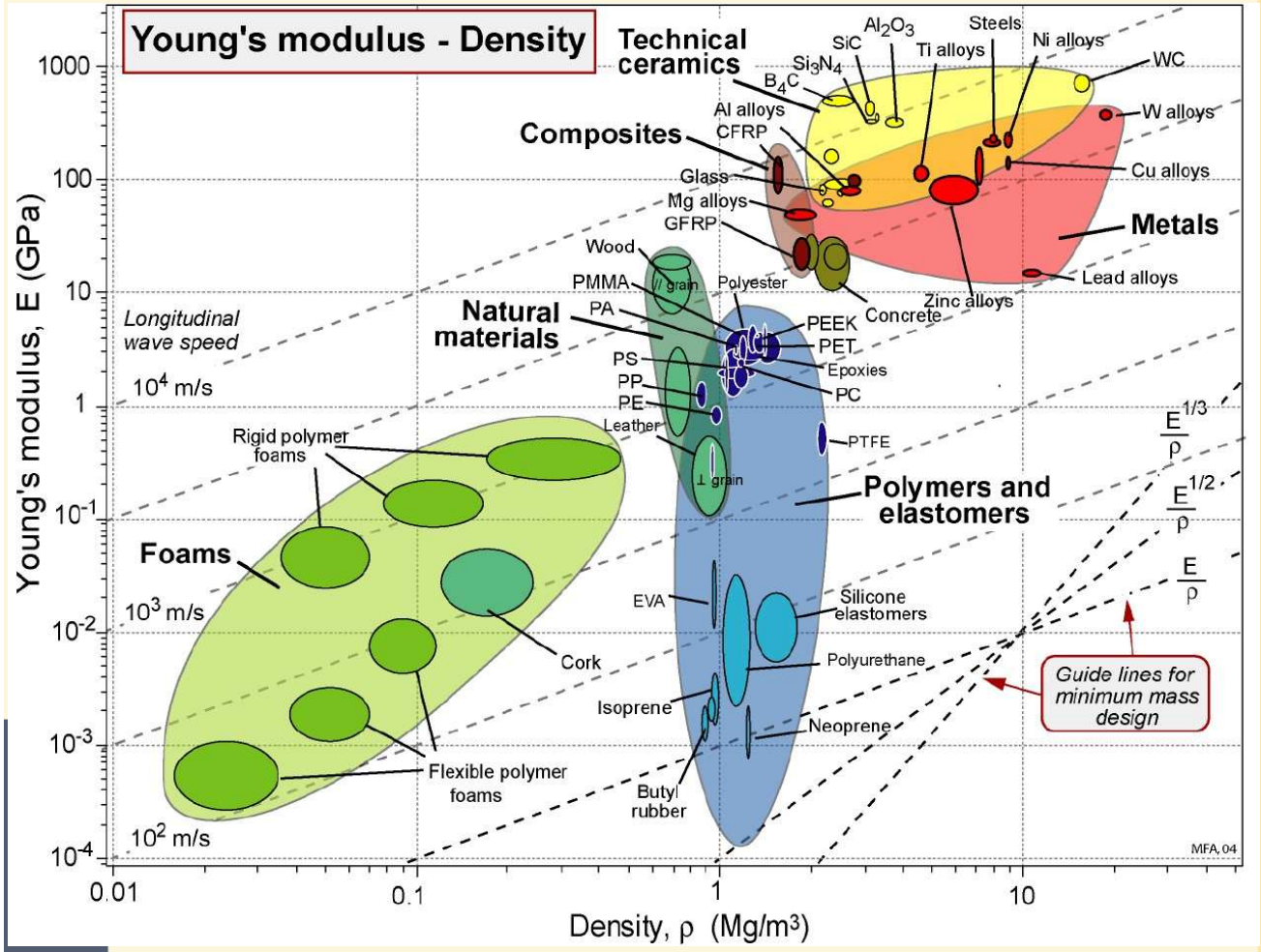
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Why do we need a selection methodology?

- Material properties vary enormously:
  - e.g. Order of magnitude difference between weakest and strongest ferrous alloys
- Multiple factors complicate design choices:
  - **Function** – loading, environment, design life etc.
  - **Cost** – availability of raw materials and manufacturing processes
  - **Time** – production deadlines and required manufacturing volumes



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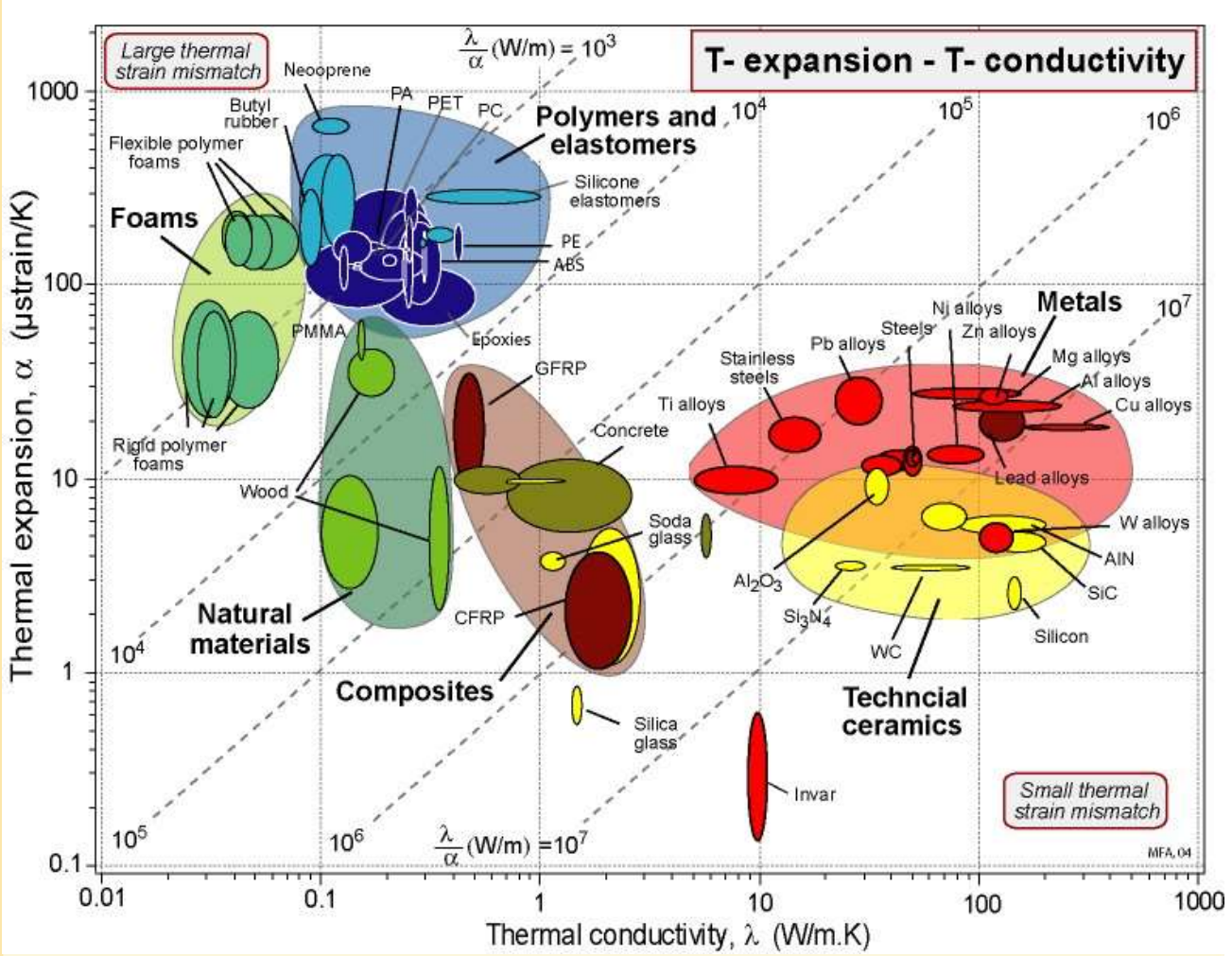
Mechanical properties

- Influencing factors:
  - Atoms
  - Bonds
  - Crystals
  - Grain structure
- Manipulated by:
  - Heat treatments (alloys)
  - Making composites
  - Making foams

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Thermal properties

- Influencing factors:
  - Bond types
  - Thermal energy is atomic vibration; propagates as waves, scattered by obstacles
- Manipulated by:
  - Composition (purity)
  - Structure





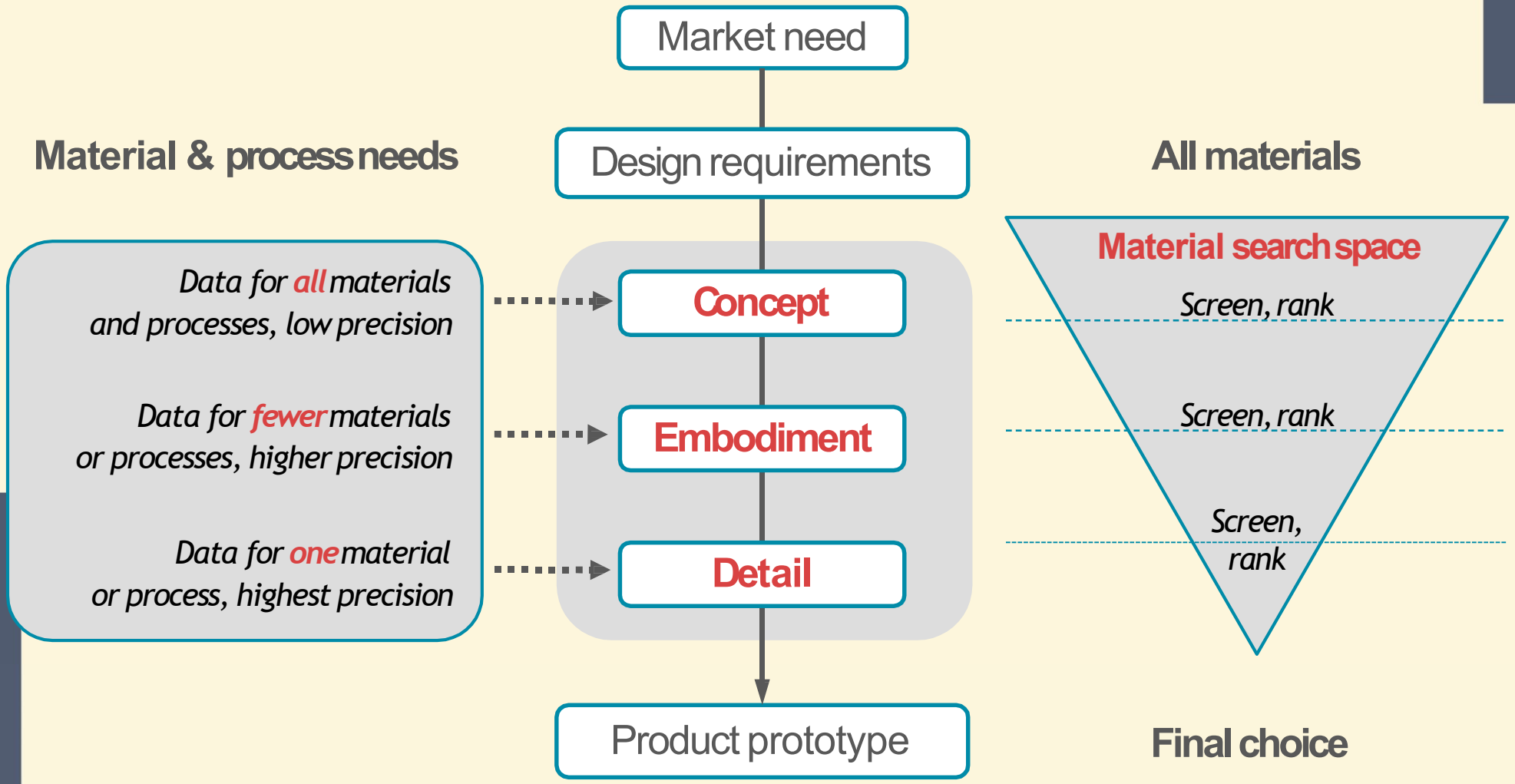
Griff on the scales

https://www.flickr.com/photos/castaway_in_wales/6182582594/

Trade-offs must be made!

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Materials selection in design



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Materials selection process (1)

1 – Translation	• Express design requirements as **constraints** and **objectives**
2 – Screening	• **Eliminate materials** based on constraints and objectives
3 – Ranking	• **Rank materials against metrics** describing best performance
4 – Supporting information	• Use additional information to **refine selection**